

MATH 220/199 Week 9 Day 1

Reminders

1. Office hours are held in Davenport 330 at 3pm on Thursdays.
2. Do the Merit Quiz on the Merit Canvas page.

The Assumptions of the MVT

First, list the 2 assumptions/criteria which must be met in order to apply the MVT.

For each of the functions and intervals below, state whether all the assumptions of the MVT are met. If an assumption is not met, list it. If all assumptions are met, state the conclusion which you get from the MVT.

1. $f(x) = |x - 3|^2$

(a) $[4, 6]$

(b) $[1, 4]$

2. $g(x) = \sqrt{x^2 - 1}$

(a) $[-2, 2]$

(b) $[3, 5]$

Exam Style Practice

1. Over the interval $[0, 3\pi]$, determine the coordinate(s) where $f(x) = 3\cos(\frac{x}{2})$ attains its minimum value.

2. Let $f(x) = \sec^2(x)$.

- (a) Over the open interval $(-\pi, \pi)$, determine the intervals where $f(x)$ is decreasing and increasing (you may present your answer in the form of a sign chart).

- (b) Determine if $f(x)$ attains maximum/minimum values over $(-\pi, \pi)$. If it does, list them.

3. Let $g(x) = \begin{cases} (x-1)^2 + 1 & x \neq 1 \\ 5 & x = 1 \end{cases}$. Determine the minimum and maximum values attained by $g(x)$ over the interval $[0, 3]$ as well as where they occur.